PROBLEM 16

Quadratic Drag

Drag on Falling Football

Original statement · Problems.pdf, p. 9

A football has frontal surface area S, mass m, and a drag coefficient C_D. It falls vertically through the air of density ρ. It is assumed that the forces acting on the ball are only gravity and drag. The drag can be modeled as: $D = \frac{1}{2}\rho v^2 S C_D$, with v the speed of the ball. Compute its terminal speed and its position as a function of time.

You may find the following integrals useful:

$$\int \frac{dx}{x_0^2 - x^2} = \frac{1}{x_0}\operatorname{atanh}\left(\frac{x}{x_0}\right); \qquad \int \tanh x \, dx = \ln \cosh x$$

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A football of mass m, frontal cross-sectional area S, and aerodynamic drag coefficient C_D falls vertically through air of density ρ.
- **Forces:**

1. Gravity: $W = mg$ downwards.

2. Aerodynamic Quadratic Drag: $D = \frac{1}{2}\rho v^2 S C_D$ upwards (opposing downward velocity $v > 0$).

- **Initial Conditions:** Released from rest at the coordinate origin:

$$s(0) = 0, \qquad v(0) = 0$$

- **Provided Integration Identities:**

$$\int \frac{dx}{x_0^2 - x^2} = \frac{1}{x_0}\operatorname{atanh}\left(\frac{x}{x_0}\right), \qquad \int \tanh x \, dx = \ln \cosh x$$

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- **Frame:** Fixed inertial frame with axis Os pointing vertically downwards.
- **Equation of Motion:**

$$m\frac{dv}{dt} = mg - \frac{1}{2}\rho S C_D v^2$$

PHASE 3

Step-by-Step Mathematical Deduction

1. Terminal Velocity v_t

> **Pedagogical Justification:** When drag equals weight, the net vertical force is zero, producing steady downward descent at constant terminal velocity v_t.

$$mg = \frac{1}{2}\rho S C_D v_t^2 \implies v_t = \sqrt{\frac{2mg}{\rho S C_D}}$$

2. Velocity Evolution $v(t)$

> **Pedagogical Justification:** Expressing the drag parameter in terms of v_t simplifies the differential equation and allows direct integration via the inverse hyperbolic tangent.

From the terminal velocity relation:

$$\frac{1}{2}\rho S C_D = \frac{mg}{v_t^2}$$

Substitute this into the equation of motion:

$$m\frac{dv}{dt} = mg - \frac{mg}{v_t^2}v^2 = mg\left(1 - \frac{v^2}{v_t^2}\right) = \frac{mg}{v_t^2}(v_t^2 - v^2)$$

Divide by m:

$$\frac{dv}{dt} = \frac{g}{v_t^2}(v_t^2 - v^2)$$

Separate variables:

$$\frac{dv}{v_t^2 - v^2} = \frac{g}{v_t^2}\,dt$$

Integrate from $v = 0$ at $t = 0$ to $v(t)$:

$$\int_0^{v(t)} \frac{dv'}{v_t^2 - v'^2} = \frac{g}{v_t^2}\int_0^t dt'$$

Apply the provided integral identity $\int \frac{dv'}{v_t^2 - v'^2} = \frac{1}{v_t}\operatorname{atanh}\left(\frac{v'}{v_t}\right)$:

$$\left[\frac{1}{v_t}\operatorname{atanh}\left(\frac{v'}{v_t}\right)\right]_0^{v(t)} = \frac{g}{v_t^2}\,t$$

Since $\operatorname{atanh}(0) = 0$:

$$\frac{1}{v_t}\operatorname{atanh}\left(\frac{v(t)}{v_t}\right) = \frac{g}{v_t^2}\,t \implies \operatorname{atanh}\left(\frac{v(t)}{v_t}\right) = \frac{gt}{v_t}$$

Take the hyperbolic tangent of both sides:

$$v(t) = v_t \tanh\left(\frac{gt}{v_t}\right)$$

3. Position Evolution $s(t)$

Pedagogical Justification: Position is found by integrating velocity over time, using the change of variable $u = \frac{gt'}{v_t}$ and the provided integral identity for $\tanh u$.

$$s(t) = \int_0^t v(t')\,dt' = \int_0^t v_t \tanh\left(\frac{gt'}{v_t}\right) dt'$$

Change of variable:

$$u = \frac{gt'}{v_t} \implies du = \frac{g}{v_t}\,dt' \implies dt' = \frac{v_t}{g}\,du$$

Limits: at $t' = 0$, $u = 0$; at $t' = t$, $u = \frac{gt}{v_t}$.

$$s(t) = v_t \cdot \frac{v_t}{g} \int_0^{gt/v_t} \tanh u \, du = \frac{v_t^2}{g} [\ln \cosh u]_0^{gt/v_t}$$

Since $\cosh(0) = 1$ and $\ln(1) = 0$:

$$s(t) = \frac{v_t^2}{g} \ln \left[\cosh \left(\frac{gt}{v_t} \right) \right]$$

This gives the exact analytical position.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Early Stages ($t \ll v_t/g$):**

Taylor expand $\tanh(u) \approx u - \frac{u^3}{3}$:

$$v(t) \approx v_t \left(\frac{gt}{v_t} \right) = gt$$

For position, $\cosh(u) \approx 1 + \frac{u^2}{2} \implies \ln \cosh(u) \approx \ln(1 + u^2/2) \approx \frac{u^2}{2}$:

$$s(t) \approx \frac{v_t^2}{g} \left(\frac{1}{2} \frac{g^2 t^2}{v_t^2} \right) = \frac{1}{2} g t^2$$

Air resistance is negligible initially; the ball is in pure gravitational free fall.

- **Late Stages ($t \gg v_t/g$):**

As $u \to \infty$, $\tanh(u) \to 1$, so $v(t) \to v_t$.

Furthermore, $\cosh(u) = \frac{e^u + e^{-u}}{2} \approx \frac{1}{2} e^u$:

$$\ln\cosh(u) \approx u - \ln 2 = \frac{gt}{v_t} - \ln 2$$

$$s(t) \approx \frac{v_t^2}{g}\left(\frac{gt}{v_t} - \ln 2\right) = v_t t - \frac{v_t^2}{g}\ln 2$$

The ball reaches terminal velocity with a constant asymptotic positional delay.

- **Dimensional check:** $[v_t] = \sqrt{[mg]/[\rho S C_D]} = \sqrt{\dfrac{\mathrm{kg}\cdot\mathrm{m}\cdot\mathrm{s}^{-2}}{\mathrm{kg}\cdot\mathrm{m}^{-3}\cdot\mathrm{m}^2}} = \mathrm{m/s}$; the argument gt/v_t of tanh is dimensionless; $[v_t^2/g] = \mathrm{m}$, so $s(t)$ is a length.
- **Numerical check (RK4, step 10^{-4}, units with $g = v_t = 1$):** integrating $\dot v = 1 - v^2$, $\dot s = v$ up to $t = 1$ gives $v = 0.761594$ and $s = 0.433781$, equal to $\tanh 1 = 0.761594$ and $\ln\cosh 1 = 0.433781$.

Source: Problems.pdf (Page 9)